

Formal Technical Referee Report

On the manuscript

“Can Agentic AI Settle a 20-Year-Old Description Logic Problem?”

“On the Decidability of $\mathcal{ALCI}_{\text{RCC5}}$, a New Decidable Fragment, and a Certified Lean Formalization Reducing It to One Keystone”

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Assessment type: Correctness, significance, and publication readiness

Overall recommendation: **Reject in the present form and invite a focused, substantially revised resubmission.**

The project contains potentially publishable mathematics, especially the claimed decidability of the $\forall\text{PO}$ -free fragment. The current manuscript, however, includes a definite mathematical error in a central one-point-extension example, overstates what has been formally certified, and does not provide reviewable proofs of its principal unconditional result or of the bridge from the live-width hypothesis F6 to complete finite certificates.

Scope note. This report assesses the supplied PDF and TeX manuscript. It does not constitute an independent build or line-by-line audit of the linked Lean repository, Python programs, mutable GitHub documents, or the manuscript’s July 2026 literature search. Claims depending on those external materials therefore remain artifact claims to be checked separately. References below to pages, sections, propositions, and theorems are references to the submitted manuscript.

Contents

1	Executive assessment	1
2	A definite mathematical error: the p. 16 extension is satisfiable	2
2.1	Why this is structurally important	3
2.2	Required repair	3
3	Major proof obligations not discharged in the manuscript	4
3.1	F6 and live width are not yet formal enough	4
3.2	The $\forall\text{PO}$ -free decidability theorem needs a full proof	5
3.3	The split forest must be separated from the strong-EQ model	6
3.4	The normal-form biconditional and amalgamation claims are overstated	7
3.5	The relationship with prior semantics needs an exact statement	8
4	TBox internalization and the recursive domino notation	8
4.1	Correction: finite TBoxes are internalizable in the full language	8
4.2	A separate mismatch remains in Section 6.2	9
5	Further technical corrections and qualifications	10
5.1	Baseline complexity of ordinary $\mathcal{ALCI}$	10
5.2	The arithmetical-hierarchy conclusion proves membership, not hardness	10
5.3	Nominals and the downarrow binder are conflated	10
5.4	Lemma 3 has an equality-labelled self-loop unless the alphabet is restricted	11
5.5	A finite generator loop is not an RCC5 model loop	11
5.6	Terminology for the implementation	11
6	Formalization and artifact claims	12
7	Substantiality, novelty, and a viable publication path	12
8	Required revisions in priority order	13
9	Final recommendation	14

1 Executive assessment

The manuscript attacks a long-standing and technically interesting decision problem: concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$ under abstract RCC5 composition-table semantics with strong equality. It combines a proposed split-forest semantics, finite-certificate ideas, a conditional decidability theorem, a claimed unconditional decidable fragment, a partial Lean formalization, computational experiments, and a methodological discussion of AI-assisted research.

There is enough potential substance for publication. In particular, a correct and fully proved theorem establishing decidability of the $\forall\text{PO}$ -free fragment, together with a precise ordered-disjoint characterization of abstract RCC5 networks and a reproducible formal artifact, would make a worthwhile specialized contribution to description logic, modal logic, or automated reasoning. The manuscript also has genuine expository value as an unusually candid record of failed proof routes.

The present version is nevertheless not ready for peer-reviewed publication as a theorem paper. The principal reasons are these:

1. The one-point-extension obstruction on p. 16 is false. The alleged contradictory requirements intersect to the single relation PO , and the extension has a simple concrete set model.
2. The false example exposes a broader invalid inference used repeatedly in the paper: although no *single* proper-relation composition cell has singleton value $\{\text{PO}\}$, several simultaneous composition constraints can force PO .
3. F6, live width, shadows, interfaces, and presentations are not defined with enough precision to support the advertised reduction. The crucial extraction theorem from bounded live width to an accepted finite certificate is not stated and proved in reviewable form.
4. The claimed unconditional $\forall\text{PO}$ -free decidability theorem is asserted with a bound but not proved in the manuscript. Its informal rationale relies on the same simplistic treatment of PO -forcing that the p. 16 counterexample refutes.
5. The split-forest construction conflates an occurrence cover with an RCC5 model. Under strong equality, distinct split copies cannot literally be “EQ-mates.” The occurrence structure, quotient, congruence conditions, and truth-preservation argument must be formally separated.
6. The normal-form biconditional is advertised as certified even though the manuscript reports Lean certification only for the forward direction; the converse is merely tested up to four elements.
7. Several further claims require correction or qualification: the ordinary $\mathcal{ALCI}$ baseline complexity, the arithmetical-hierarchy language, the nominal/binder distinction, the path-automaton cycle statement, and the use of “decision procedure” for an empirically tested but unproved reasoner.

Criterion	Assessment	Present status
Originality	The combination of split presentations, RCC5 local algebra, and AI-assisted formal development is interesting. Novelty of individual algebraic results still needs a careful literature check.	Promising
Significance	Potentially high if the $\forall\text{PO}$ -free theorem and normal-form/amalgamation results are correct and fully proved.	Potentially publishable
Correctness	At least one definite central error; several load-bearing claims lack reviewable proofs.	Not yet adequate
Clarity	Broadly readable, but technical, historical, software, and advocacy narratives compete with one another.	Major refocusing needed
Reproducibility	The repository and Lean claims are promising, but the paper does not provide a stable, self-contained artifact contract.	Requires artifact audit

2 A definite mathematical error: the p. 16 extension is satisfiable

The manuscript presents the following as a minimal example showing that a composition-closed old network may fail to admit a requested fresh witness. The old points satisfy

$$a \text{ DR } b, \quad a \text{ PPI } c, \quad b \text{ DR } c,$$

and the new point d is required to satisfy

$$b \text{ PO } d, \quad c \text{ PO } d.$$

The manuscript claims that the two relevant triangles force both $a \text{ DR } d$ and $a \text{ PPI } d$.

That calculation is incorrect. In the manuscript's convention for the RCC5 table,

$$\text{comp}(\text{DR}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\} \quad \text{and} \quad \text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}, \text{PPI}\}.$$

Therefore the relation from a to d must belong to

$$\{\text{DR}, \text{PO}, \text{PP}\} \cap \{\text{PO}, \text{PPI}\} = \{\text{PO}\}.$$

The two triangles force $a \text{ PO } d$; they do not create a contradiction.

A concrete set model makes the point unambiguous. Over the universe $\{1, 2, 3, 4, 5, 6\}$, let

$$a = \{1, 2, 6\}, \quad b = \{3, 5\}, \quad c = \{1, 2\}, \quad d = \{1, 3, 4\}.$$

Using the standard RCC5 relations on sets, one has

$$a \text{ DR } b, \quad a \text{ PPI } c, \quad b \text{ DR } c, \quad b \text{ PO } d, \quad c \text{ PO } d, \quad a \text{ PO } d.$$

For example, $b \cap d = \{3\}$ and both $b \setminus d$ and $d \setminus b$ are nonempty, so $b \text{ PO } d$; similarly for c and d . Thus all stated old and new requirements are jointly satisfiable.

2.1 Why this is structurally important

Proposition 1 itself makes the narrower and correct claim that no *single* composition of two proper RCC5 relations returns the singleton $\{\text{PO}\}$. The manuscript repeatedly strengthens this to claims such as “PO is never forced,” “composition can never determine a horizontal relation,” and “the steering function is the one genuine degree of freedom.” Those stronger claims are false without qualification.

In a complete atomic network, the label of an edge must satisfy the constraints induced by *all* relevant intermediate points. Intersections of non-singleton table entries may be singletons, as the example above demonstrates. A relation can also become unique after concept-theoretic filtering or after closure under other constraints. The correct distinction is therefore:

No single proper-relation length-two path forces PO, but the intersection of constraints generated by several paths may force PO.

This distinction affects every later use of the notions “shadow,” “live edge,” “free horizontal choice,” and “horizontal residue.” In particular, an edge can be a shadow because its value is fixed only by the *joint closure* of several constraints, not by one determining table entry. Any width count based on single-path forcing can overcount or undercount the live information.

2.2 Required repair

The paper should perform a systematic audit of each use of “PO is never forced” and distinguish at least four notions:

1. forcing by one length-two path;
2. forcing by intersection over several intermediate points;
3. forcing after universal-concept or type filtering;

4. forcing after transitive, converse, equality, and disjointness closure.

The p. 16 example must be removed or replaced by a correct non-extendibility example. Since the manuscript later claims an “exact one-point extension criterion,” that criterion should be stated and used to certify the replacement example.

3 Major proof obligations not discharged in the manuscript

3.1 F6 and live width are not yet formal enough

Definition 4 is intuitive but not adequate as the premise of a mathematical theorem. At least the following terms remain underdefined:

- the class of admissible “presentations” of a model;
- the mathematical object called an “interface”;
- the order, if any, in which edges are regarded as “already present”;
- the closure operator used to decide whether an edge is forced;
- whether forcing is single-path, finite-path, or global finite closure;
- whether live width is measured for one presentation, minimized over presentations, or minimized over presentation orders;
- the finite encoding of a presentation and the effective relationship between width and certificate size.

The error in Section 2 makes these omissions consequential rather than merely stylistic. If an edge is forced by the intersection of two constraints, it should normally count as a shadow, but Definition 4’s phrase “forced by composition from edges already present” does not say whether that joint inference is included.

The conditional decision theorem should be decomposed into explicit statements. A standard target would be an equivalence of the form

$$C \text{ is satisfiable} \iff \exists K \in \text{Cert}_{\leq f(C)} \text{ Check}(C, K),$$

where:

1. $\text{Cert}_{\leq f(C)}$ is an effectively enumerable finite set;
2. Check is a total recursive predicate;
3. accepted certificates unfold to genuine strong-EQ RCC5 models (soundness);

4. every satisfiable input admits an accepted certificate within the bound (completeness/extraction).

The manuscript reports machine checking of item 3. The mathematically difficult step is item 4, including the passage from live-width presentations to the particular finite certificate syntax and the $W2'$ uniformization conditions. That bridge is not formally stated and proved in the paper.

If F6 is defined so strongly that it already says every satisfiable formula has an accepted certificate in an effectively bounded search space, the conditional decidability theorem becomes a direct enumeration corollary. If F6 is intended to be a weaker semantic width property, then the bridge from width to certificates is itself a substantial theorem and must be supplied.

The claimed equivalence

$$\text{F6 fails} \iff \text{some concept forces unbounded identity-selector minors}$$

is likewise not reviewable from the current manuscript. “Identity-selector minor,” “forces,” “separator cover,” and the two directions of the equivalence require formal definitions and proofs, not only repository pointers and work-package names.

Finally, the phrase “F6 is the open problem itself” is logically too strong. From the advertised theorem one obtains

$$\text{F6} \implies \text{decidability},$$

and hence

$$\text{undecidability} \implies \neg \text{F6}.$$

Failure of F6 would not by itself prove undecidability: the logic might be decidable by a representation not controlled by this width measure. In particular, a concept forcing unbounded identity-selector minors would refute the proposed certificate route, but would not prove undecidability without an additional effective reduction. Nor must every undecidability proof take the form of a rigid two-dimensional grid. F6 may be the keystone of this proof architecture without being equivalent to the decision problem.

3.2 The $\forall\text{PO}$ -free decidability theorem needs a full proof

Theorem 9 is potentially the manuscript’s strongest unconditional result. It states decidability of the $\forall\text{PO}$ -free fragment with the explicit bound

$$K(C_0) = (1 + m)^2 2^{2^m}, \quad m = |\text{cl}(C_0)|.$$

The manuscript gives only a short informal motivation and pointers to repository directories. A referee cannot verify from the paper:

- the exact two-tier quotient construction;

- the definition of phases, kernels, designated witnesses, and their equivalence relation;
- preservation of existential restrictions;
- preservation of $\forall\text{DR}$, $\forall\text{PP}$, and $\forall\text{PPI}$ under quotienting and unfolding;
- strong-EQ preservation and representative independence;
- the derivation of the displayed bound;
- termination and correctness of the resulting decision algorithm.

The informal reason given for the fragment is that PO is the unique relation needing an “all-phase” condition. That intuition must be re-audited after the p. 16 error: PO can be fixed by the intersection of several path constraints even when no single table cell determines it. This does not show that Theorem 9 is false, but it removes confidence in any proof that treats all PO edges as freely chosen or inherently non-shadow.

The theorem’s scope should also be stated carefully. It concerns bare concepts with no $\forall\text{PO}$ subformula after the chosen normal-form transformation. General TBox internalization in the full language, discussed in Section 4 below, introduces a universal modality containing a $\forall\text{PO}$ conjunct. Thus arbitrary TBoxes are internalizable in full $\mathcal{ALCT}_{\text{RCC5}}$, but the internalization does not in general preserve the $\forall\text{PO}$ -free fragment.

A revised technical paper should place a conventional, self-contained proof of Theorem 9 in the main text or in a stable formal appendix. If the proof is too long, the paper should state every lemma precisely and point to one archived companion proof, not to a collection of mutable development folders.

3.3 The split forest must be separated from the strong-EQ model

The manuscript says that a point lying under several incomparable superparts is split into “EQ-mates,” after which the mates are re-identified by a typed-EQ quotient. Under the manuscript’s strong-EQ semantics,

$$x \text{ EQ } y \iff x = y.$$

Distinct split occurrences therefore cannot literally be EQ-related elements of an intermediate RCC5 model.

The intended construction appears to be an *occurrence cover*: a forest of occurrences equipped with an external equivalence relation, followed by a quotient whose equivalence classes are the actual model points. That is a legitimate strategy, but the paper must formally distinguish:

1. the occurrence domain;
2. the external equivalence relation on occurrences;

3. the forest or cover relation on occurrences;
4. the RCC5 relation on equivalence classes;
5. atomic concept labels on occurrences and their coherence on classes;
6. satisfaction in the cover versus satisfaction in the quotient.

For the quotient to be well defined, the RCC5 label between two equivalence classes must not depend on the chosen representatives. Existential witnesses, universal restrictions, converse, composition closure, and strong equality must all survive quotienting. Conversely, the completeness direction must show how every model yields such a forest cover without changing truth at the distinguished point.

The manuscript currently presents the split-forest normal form as a robust semantic foundation, but does not give a theorem with exact hypotheses, construction, and proof. A whiteboard diagram and the fact that earlier cold reviews did not refute the idea are evidence about research process, not a substitute for this theorem.

3.4 The normal-form biconditional and amalgamation claims are overstated

Theorem 12 is stated as an if-and-only-if characterization of composition-closed strong-EQ RCC5 networks by a strict partial order, hereditary disjointness, and residual PO. The text then says that only the forward direction is proved in Lean, while the converse is exhaustively tested up to size four.

Accordingly, the manuscript has not established a *machine-certified biconditional*. The accurate description is:

The forward characterization is Lean-certified for arbitrary domains; the converse is conjectured or proved informally and has been exhaustively tested on structures of size at most four.

The converse looks plausible and may admit a concise case analysis. That makes completing it especially important. Finite testing through four elements is not a proof for arbitrary structures, and the later claims of free amalgamation and exact one-point extension depend on the converse direction.

The use of “patchwork property” also needs clarification. Under the paper’s abstract semantics, a complete atomic network whose every triangle satisfies the composition table is already a model by definition. In that setting, saying that triple-wise consistency implies global consistency is close to definitional. What the split-forest argument appears to need is a stronger extension or amalgamation theorem:

Compatible complete or partial atomic networks on overlapping domains can be combined, with suitable cross-edge labels, into a larger composition-closed network.

The paper should state the exact property required: whether networks are finite, whether they are complete or partial, what agreement is required on overlaps, whether the target is abstract table consistency or representability by actual regions, and whether the theorem supports iterated or countable patching. Empirical success on finitely many two-bag amalgamations is useful regression evidence but not a general theorem.

3.5 The relationship with prior semantics needs an exact statement

The manuscript identifies $\mathcal{ALCT}_{\text{RCC5}}$ with the logic denoted $L_{\text{RCC5}}(\text{RS})$ in earlier work, while nearby prose discusses semantics over regular closed regions in topological spaces. Since the present paper deliberately uses arbitrary complete edge-labelled structures satisfying only the RCC5 table, the claimed equivalence is not self-evident.

This point underpins both the asserted 20-year open-problem status and the novelty claim. A revised paper should include a proposition or an exact bibliographic statement specifying the class denoted RS, the role of abstract versus concrete region structures, and why the two satisfiability problems coincide. The same care is needed for the manuscript's broad claims about all standard decidable first-order fragments and the current literature status.

4 TBox internalization and the recursive domino notation

4.1 Correction: finite TBoxes are internalizable in the full language

The manuscript's concept syntax contains no primitive TBox or fixed-point constructor, but the five RCC5 relations form a total partition of the square of the domain. Consequently the full language has a definable universal modality. For any concept D , define

$$[U]D := \forall \text{EQ}.D \sqcap \forall \text{PP}.D \sqcap \forall \text{PPI}.D \sqcap \forall \text{PO}.D \sqcap \forall \text{DR}.D.$$

For every model and every point x ,

$$x \in ([U]D)^{\mathcal{I}} \iff D^{\mathcal{I}} = \Delta^{\mathcal{I}},$$

because every ordered pair carries exactly one RCC5 base relation. Inclusion of EQ covers the evaluation point itself.

Let a finite TBox over the same language be

$$\mathcal{T} = \{C_i \sqsubseteq D_i : 1 \leq i \leq n\},$$

and put

$$G_{\mathcal{T}} = \bigwedge_{i=1}^n (\text{nmf}(\neg C_i) \sqcup D_i).$$

Then

$$C \text{ is satisfiable with respect to } \mathcal{T} \iff C \sqcap [U]G_{\mathcal{T}} \text{ is satisfiable.}$$

Thus the use of a cyclic concept equation in the domino discussion is not, by itself, an expressivity error. If $F(X)$ is the displayed right-hand side and X is an atomic concept name, the descriptive TBox axiom

$$X \equiv F(X)$$

can be internalized, for the purpose of asking for an X -point, as

$$X \sqcap [U]((\neg X \sqcup F(X)) \sqcap (\text{nmf}(\neg F(X)) \sqcup X)).$$

If only the unfolding direction $X \sqsubseteq F(X)$ is intended, the second implication is omitted.

The manuscript should add this observation explicitly. It resolves the apparent syntax/TBox mismatch in the full language. As noted above, the internalization uses $\forall\text{PO}$, so it does not preserve the $\forall\text{PO}$ -free fragment.

4.2 A separate mismatch remains in Section 6.2

Internalization does not fix the inconsistency between the formula displayed, the prose argument, and the computational test. The displayed recursive conjunct is

$$\forall\text{PPI}.\exists\text{PPI}.X.$$

The prose then analyzes “adding $\forall\text{PPI}.X$ ” and says that a PPI-child must itself satisfy X . The reported finite experiment tests

$$X_2 = X_1 \sqcap \forall\text{PPI}.X_1.$$

These are different formulas. Under $\forall\text{PPI}.\exists\text{PPI}.X$, a PPI-successor need only have a further PPI-successor satisfying X ; it need not itself satisfy X . Consequently the two-level contradiction described in the prose does not establish failure of the displayed recursive schema.

The authors should choose one construction and align all three components:

1. the recursive TBox or internalized concept;
2. the semantic explanation of the next unfolding;
3. the formula actually submitted to the computational checker.

At present, the experiment verifies a different and apparently stronger recurrence than the one

printed.

5 Further technical corrections and qualifications

5.1 Baseline complexity of ordinary $\mathcal{ALCI}$

Table 1 labels ordinary $\mathcal{ALCI}$ concept satisfiability as ExpTime-complete. For bare concept satisfiability without general TBoxes, the standard complexity is PSpace-complete; ExpTime-completeness is the standard result with general TBoxes or GCIs. The table heading says “concept satisfiability,” so the row is misleading unless it is relabelled.

The TBox internalization above is specific to the total RCC5 role partition in the full logic. It does not retroactively give ordinary $\mathcal{ALCI}$, with unconstrained role names and no universal role, the same internalization. The paper should therefore state consistently whether each complexity entry concerns bare concepts, concepts with general TBoxes, or a language with a universal modality.

5.2 The arithmetical-hierarchy conclusion proves membership, not hardness

The first-order translation shows that satisfiable $\mathcal{ALCI}_{\text{RCC5}}$ concepts form a co-r.e. set, hence belong to Π_1^0 . It does not show Π_1^0 -hardness or completeness for this particular problem. A decidable set also belongs to Π_1^0 . Phrases such as “the same level as the domino problem” and “the knife’s edge is literal” should therefore be replaced by the precise claim:

Satisfiability is in Π_1^0 ; no matching lower bound is established here.

The dovetailing observation is sound in its proper form: if accepted finite certificates recursively enumerate all satisfiable inputs, while first-order refutations recursively enumerate all unsatisfiable inputs, then the problem is decidable. This is a standard semidecidability argument. It does not independently validate the proposed certificate class.

5.3 Nominals and the downarrow binder are conflated

The preliminaries correctly observe that singleton concepts are definable under strong equality. An atomic concept i together with

$$i \sqcap \bigwedge_{R \in \{\text{PP}, \text{PPI}, \text{PO}, \text{DR}\}} \forall R. \neg i$$

forces i to hold at exactly one point in the relevant model. Later, the paper attributes the absence of counterexamples to “the absence of nominals in $\mathcal{ALCI}$.” These claims are inconsistent.

The feature actually missing appears to be a hybrid downarrow binder or another mechanism for dynamically naming the current point and reusing that name inside nested modalities. A nominal

and a binder are not the same expressive resource. The paper should use the hybrid-logic terminology precisely.

5.4 Lemma 3 has an equality-labelled self-loop unless the alphabet is restricted

The subset automaton is initialized with $\text{Rel}(\varepsilon) = \{\text{EQ}\}$, and EQ is the identity for composition. Hence, for every reachable state S ,

$$\text{Rel}(w\text{EQ}) = \text{Rel}(w),$$

so there is an EQ-labelled self-loop. The literal statement that every cycle uses only vertical labels is false unless equality transitions are excluded by definition or equality is explicitly included in the benign class.

The likely intended statement is that no cycle contains a DR- or PO-labelled transition. The manuscript should state the alphabet restriction and the exact machine-checked proposition.

5.5 A finite generator loop is not an RCC5 model loop

The introduction says that an unbounded vertical chain “folds into a finite loop.” In a strong-EQ RCC5 model, PP is strict and a PP-cycle is impossible. Later the paper gives the correct regular-cover interpretation: a finite generator may contain a self-loop whose unfolding creates infinitely many distinct points.

The introductory wording should be changed accordingly. The distinction between a finite presentation and the model it unfolds is central to the paper and should not be blurred at the outset.

5.6 Terminology for the implementation

In mathematical logic, a “decision procedure” normally denotes an algorithm proved to terminate and return the correct answer on every input. The manuscript explicitly says that no completeness theorem is known for the cover-tree tableau. It should therefore be called a prototype reasoner, candidate procedure, or experimental tableau.

The empirical evidence remains valuable, but its scope should be stated accurately:

- a finite model verifier can independently validate explicit finite SAT witnesses;
- it cannot validate UNSAT answers;
- it cannot validate SAT answers requiring infinite models unless the reasoner emits a separately verified regular certificate;

- the two implementations share the same project, relation table, and test infrastructure, leaving common-mode risk;
- the cycle-aware wrapper was introduced to repair a known blind spot, so the 911-case agreement is not evidence from two wholly independent procedures.

These tests are excellent regression tests for the implementation and local algebra. They are not substitutes for the missing completeness proofs.

6 Formalization and artifact claims

The manuscript’s use of Lean is a strength, but the claim boundaries need to be exact. “Zero **sorry**” is not by itself a complete artifact audit. A publication-quality artifact should provide:

1. a stable archive or commit hash rather than only mutable branch paths;
2. the exact Lean version and a one-command build script;
3. the exported theorem signatures corresponding to every claim in the abstract;
4. `#print axioms` output for each load-bearing theorem;
5. a clear statement of where `native_decide` is used and what trusted computing base this adds;
6. a map from manuscript definitions to Lean definitions;
7. an explicit distinction between the certificate soundness theorem, abstraction faithfulness, completeness/extraction, and decidability.

The abstract and contribution list should not call the RCC5 normal-form biconditional “certified” until both directions are kernel-checked or the uncertified direction is clearly separated. Likewise, “soundness half” should not be presented in a way that readers can mistake for a machine-checked decision theorem.

The large bibliography of AI-generated technical reports is useful as a development archive, but it is not an acceptable replacement for proofs in the submitted paper. Core claims should be proved in the article or in one stable, archived companion document. Superseded drafts, conversation logs, and work-package directories may remain supplementary provenance.

7 Substantiality, novelty, and a viable publication path

The project is not insubstantial. A focused technical paper could make a meaningful contribution if it establishes the following in conventional form:

1. a precise ordered-disjoint representation theorem for abstract strong-EQ RCC5 networks, with both directions proved;
2. the exact finite or countable amalgamation property needed by the construction;
3. a formal split-occurrence cover and quotient theorem preserving concept satisfaction;
4. a complete proof of decidability for the $\forall\text{PO}$ -free fragment, including the bound and algorithm;
5. a reproducible Lean artifact tied to exact theorem statements.

A suitable technical title would be along the lines of

An Ordered-Disjoint Normal Form for Abstract RCC5 Networks and Decidability of the $\forall\text{PO}$ -Free Fragment of $\mathcal{ALCT}_{\text{RCC5}}$.

The AI-assisted methodology, cold-review process, failed routes, and lessons for proof engineering could form a separate experience or position paper. Combining a theorem paper, software report, research diary, AI manifesto, historical priority discussion, and response to a workshop rejection makes the present manuscript difficult to evaluate by any one scholarly standard.

Appendix A on the DL 2026 rejection should be removed from a technical submission. Its accusations about community motives, disclosure incentives, and bias are not relevant to the validity of the mathematical claims and are likely to distract from the strongest material. The exhortation that researchers declining to use these tools are “simply unwise” should likewise be omitted or moved to a separate opinion piece.

The title should also be reconsidered. The paper explicitly does not settle the open problem. A question-form title is not formally false, but it foregrounds a settlement that the paper does not deliver and contributes to the sense of overclaiming.

8 Required revisions in priority order

Before the manuscript is resubmitted as a technical paper, I recommend the following sequence.

1. **Correct the p. 16 extension calculation** and audit every use of “PO is never forced” for joint path-constraint intersection.
2. **State a formal closure notion for shadows and live edges**, including presentation order, interfaces, and minimization conventions.
3. **Give a complete proof of Theorem 9** in the paper or a single stable companion artifact; re-check it against multi-path PO forcing.

4. **Formalize the split-forest occurrence cover and quotient**, keeping distinct occurrences separate from strong-EQ model elements.
5. **Prove both directions of the ordered-disjoint normal form** and state the exact amalgamation and one-point-extension theorems.
6. **State the F6-to-certificate extraction theorem explicitly**, with $W2'$ and identity-selector notions formally defined.
7. **Align the recursive domino formula, prose, and experiment**; add the universal-modality/TBox internalization lemma.
8. **Correct the complexity, Π_1^0 , nominal/binder, path-automaton, and finite-generator statements**.
9. **Downgrade empirical terminology**: “prototype reasoner” rather than “decision procedure,” and “tested” rather than “certified” where appropriate.
10. **Refocus the article**: remove the rejection appendix and advocacy passages, shorten the development diary, and move non-load-bearing historical material to supplementary documentation.
11. **Archive and audit the artifacts** with exact versions, theorem signatures, axioms, and build instructions.

9 Final recommendation

I recommend **rejection in the current form, with encouragement to resubmit a substantially narrower and fully proved technical paper**. This is not a judgment that the project lacks scientific value. The manuscript identifies an interesting problem, contains potentially strong partial results, and documents an ambitious formal-development effort. The $\forall$ PO-free fragment alone may warrant publication if its theorem is correct, novel, and supplied with a complete proof.

The present version, however, asks the reader to accept too many load-bearing transitions on the basis of informal descriptions, repository pointers, finite experiments, and partial certification. The false one-point-extension example further shows that the central “horizontal freedom” intuition has not yet been formulated carefully enough. The next improvement should come from formal definition, proof consolidation, and claim reduction, not from additional rounds of testing or further broad methodological claims.

Concise verdict. The work is potentially substantial and interesting, but it does not yet warrant publication as a technical theorem paper. The p. 16 error must be repaired first; the $\forall$ PO-free theorem and split-forest/F6 bridge then need complete, self-contained proofs. If those results survive that process, a focused resubmission would merit serious consideration.